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Lab1- Transients in RLC Networks

1. Analysis – Step1 (25%)

1. Write the transfer function in time domain for the voltage across C_1 versus the input voltage V_1.
   a. Convert V1 and Rth into its equivalent current source Ix and parallel resistor

Ix	$I_x = \frac{V_1}{R_{th}}$
Rparallel	$R_{parallel} = R_{th}$

   b. Combine the parallel resistor with R2 giving Req

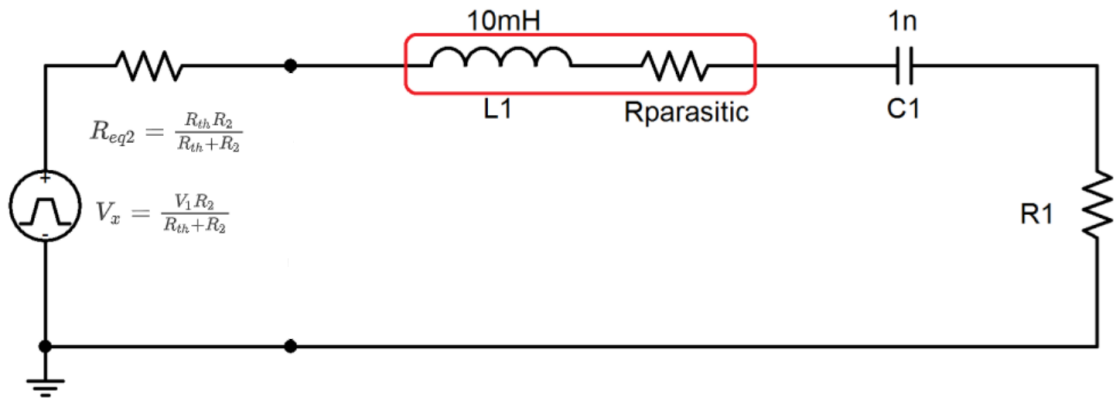
Req	$R_{eq} = \frac{R_{th}R_2}{R_{th}+R_2}$

   c. Convert Ix and Req into Vx and Req2

Vx	$V_x = \frac{V_1 R_2}{R_{th}+R_2}$
Req2	$R_{eq2} = R_{eq} = \frac{R_{th}R_2}{R_{th}+R_2}$

   d. Rearrange the components so as to get a RLC series circuit – Redraw circuit in Fig1 as instructed.

Figure of the equivalent circuit



   e. Apply mesh analysis to the RLC series

Mesh analysis
We have: $$-V_x + V_R + V_{L1} + V_{C1} = 0$$ $$-V_x + R_{eq2} \times I + L_1 \times \frac{dI}{dt} + \frac{1}{C_1}\int_{-\infty}^{\tau=t} I(\tau)d\tau = 0$$

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$$V_1 = IR_\{th\}⁄\{L_1\ (R_\{th\} + R_2)\}R_2 \times ⁄\{dI\}\{dt\}\}$$

$$V_1 = IR_{th} + \frac{L_1\ (R_{th} + R_2)}{R_2} \times \frac{dI}{dt} + \frac{R_{th} + R_2}{C_1 R_2} \int_{-\infty}^{\tau=t} I(\tau) d\tau$$

f. Give the transfer function of V_{C1} as a function of V_1 and the component values.

Transfer function

Since

$$V_1 = IR_{th} + \frac{L_1\ (R_{th} + R_2)}{R_2} \times \frac{dI}{dt} + \frac{R_{th} + R_2}{C_1 R_2} \int_{-\infty}^{\tau=t} I(\tau) d\tau$$

$$V_{C1} = \frac{1}{C_1} \int_{-\infty}^{\tau=t} I(\tau) d\tau$$

So

$$\frac{V_{C1}}{V_1} = \frac{\frac{1}{C_1} \int_{-\infty}^{\tau=t} I(\tau) d\tau}{IR_{th} + \frac{L_1\ (R_{th} + R_2)}{R_2} \times \frac{dI}{dt} + \frac{R_{th} + R_2}{C_1 R_2} \int_{-\infty}^{\tau=t} I(\tau) d\tau}$$

2. Determine the natural frequency, damping factor, and damping frequency of the circuit shown in Figure 2 where $R_{th} = 50\Omega$, $R_2 = 47\Omega$, $R_{parasitic} = 100\Omega$ and a) $R_1 = 300\Omega$ and b) $R_1 = 3k\Omega$. Note that you should combine R_{th}, $R_{parasitic}$, R_1, and R_2 into one equivalent resistor before performing these calculations. Be careful with units!!! (Hz and rad/s are not the same)

Req a)	$R_{eq} = \frac{R_{th} R_2}{R_{th} + R_2} + R_{parasitic} + R_1 = \frac{50 \times 47}{50 + 47} + 100 + 300 = 424.2\Omega$
Req b)	$R_{eq} = \frac{R_{th} R_2}{R_{th} + R_2} + R_{parasitic} + R_1 = \frac{50 \times 47}{50 + 47} + 100 + 3000 = 3124.2\Omega$
ω_0	$\omega_0 = \frac{1}{\sqrt{LC}} = 3.16 \times 10^5 rad/s$
α a)	$\alpha = \frac{R_{eq}}{2L} = 21210 rad/s$
α b)	$\alpha = \frac{R_{eq}}{2L} = 1.56 \times 10^5 rad/s$
ζ a)	$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} = \frac{424.2}{2}\sqrt{\frac{1 \times 10^{-9}}{10 \times 10^{-3}}} = 0.067$
ζ b)	$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} = \frac{3124.2}{2}\sqrt{\frac{1 \times 10^{-9}}{10 \times 10^{-3}}} = 0.49$

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ω_d a)	$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{(3.16 \times 10^5)^2 - 21210^2} = 3.15 \times 10^5 \text{rad/s}$
ω_d b)	$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{(3.16 \times 10^5)^2 - (1.56 \times 10^5)^2} = 2.75 \times 10^5 \text{rad/s}$

3. For the two cases a) $R_1 = 300\Omega$ and b) $R_1 = 3k\Omega$, determine the values for A_1 and A_2 in the equation for a forced response, $v_C(t) = [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)]e^{-\alpha t} + Vs$ when $V_1 = 2V[1 + 2u(t)]$, where $u(t)$ is a unit step function. Note that $Vs \neq V_1$. The solution will be the same as the case where $V_1 = 4Vu(t)$ because the transient response is only produced by the step function, not the d.c. component of V_1.

a) Determine the initial voltage at t = 0⁻s, just before the step input.

$v_1(0^- s)$	$v_1(0^- s) = 2V$
$v_C(0^- s)$	$v_C(0^- \text{s}) = \frac{R_2}{R_{th} + R_2} v_1(0^-) = \frac{47}{50 + 47} \times 2 = 0.97V$

b) Determine the steady state voltage at t = ∞s

$v_1(\infty s)$	6V
$v_C(\infty s)$	2.91V

c) Considering $v_C(t) = [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)]e^{-\alpha t} + Vs$ at t = ∞s, solve V_S

V_S	$V_S = v_C(\infty s) = 2.91V$

d) Considering $v_C(t) = [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)]e^{-\alpha t} + Vs$ at t = 0⁻s, solve A1

$A1$	$A_1 = -1.94V$, since $v_C(0s) = A_1 + V_S = A_1 + 2.91 = 0.97$

e) Considering $i_c(t)$ at t = 0⁻s, solve A2 (refer to Hint 2 – derivation needed – A2 depends on R1 so you should have 2 different values for both cases)

$derivation$	$i_c(t)$ $= C\frac{dv_c(t)}{dt} = [(A_2\omega_d - A_1\alpha)\cos(\omega_d t) - (A_1\omega_d + A_2\alpha)\sin(\omega_d t)]e^{-\alpha t}C$ Since $i_c(0s) = 0$, we have $A_2\omega_d - A_1\alpha = 0$, So $A_2 = \frac{\alpha}{\omega_d}A_1$

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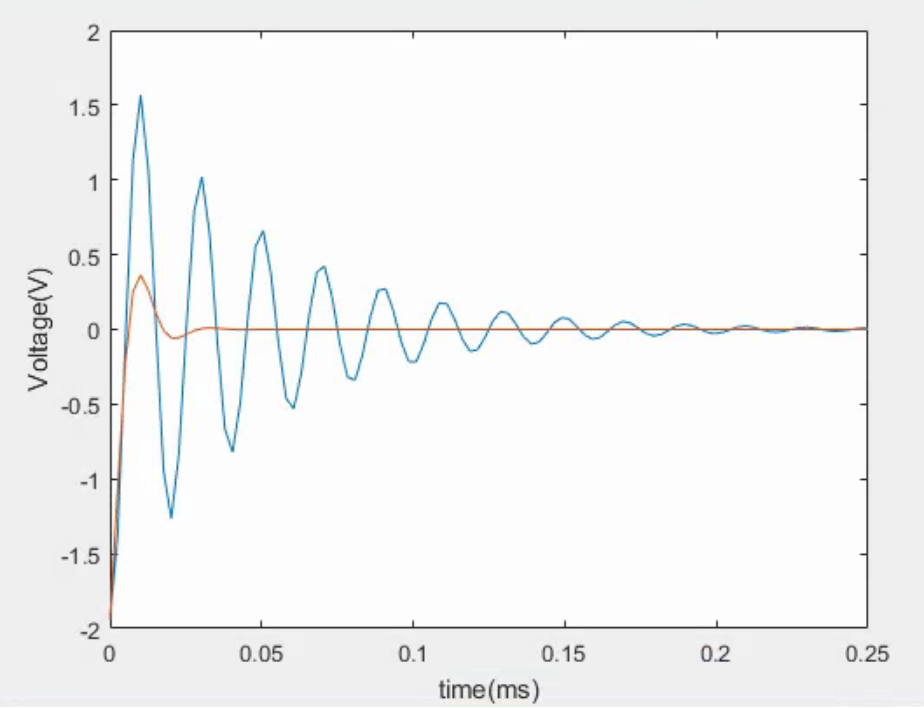
A2 a)	$A_2 = \frac{\alpha}{\omega_d} A_1 = \frac{21210}{3.15 \times 10^5} \times (-1.94) = -0.13V$
A2 b)	$A_2 = \frac{\alpha}{\omega_d} A_1 = \frac{1.56 \times 10^5}{2.75 \times 10^5} \times (-1.94) = -1.1V$

4. Using Matlab, plot the equations for the voltage across the capacitor developed in Step 3 on the same graph from t = 0 s to t = 0.25 ms.

Insert a screen capture of your code with comments

```
t = linspace(0, 0.25, 100) / 1000;
Vca = [-1.94 * cos(315000*t) - 0.13 * sin(315000*t)] .* exp(-21210*t);
Vcb = [-1.94 * cos(274000*t) - 0.13 * sin(274000*t)] .* exp(-156000*t);
plot(t, Vca, Vcb);
```

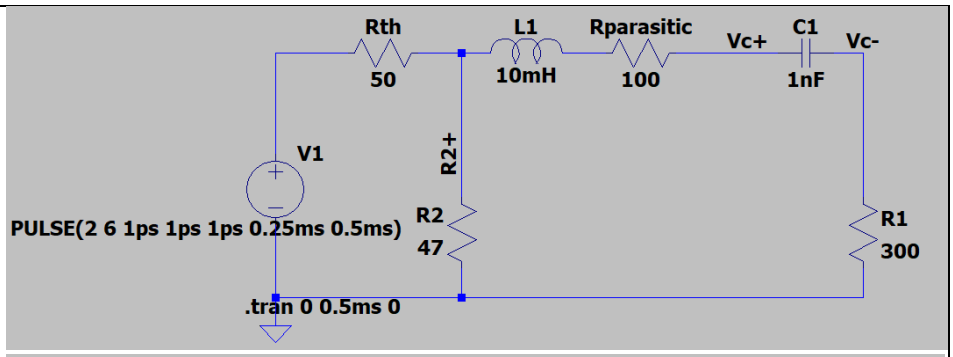
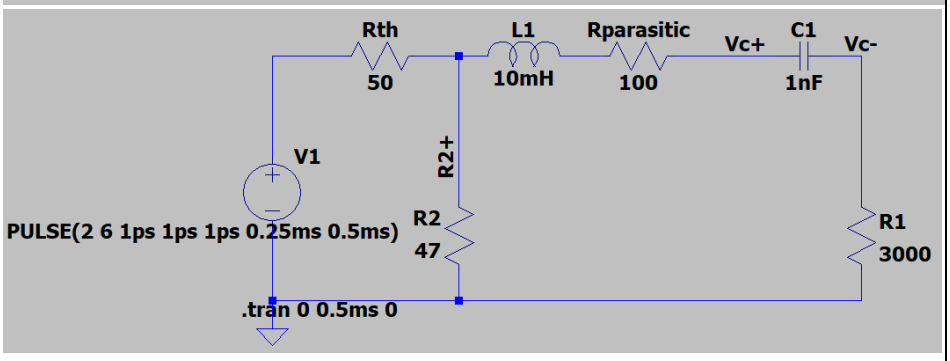
Insert the resulting graph with appropriate annotations, units and legends.



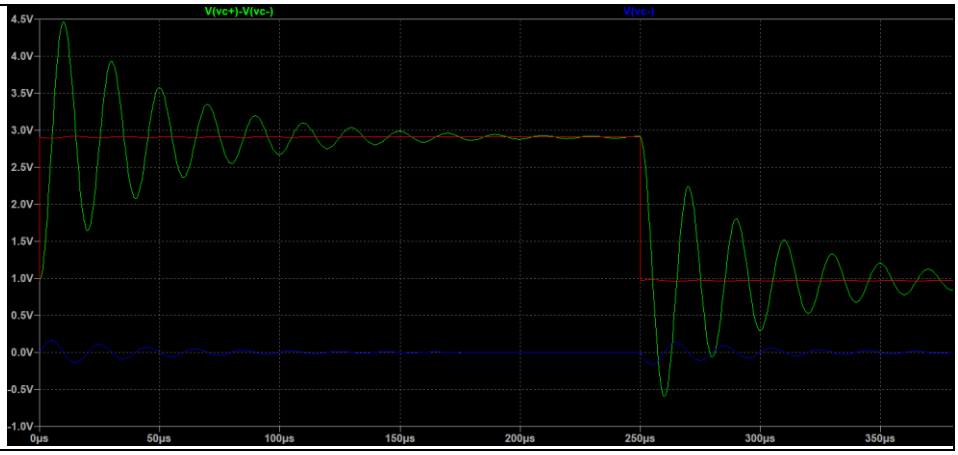
2. Modeling – Step2 (25%)

1. In PSpice, simulate the transient response of the circuit shown in Figure 2 where $R_{th} = 50\Omega$, $R_2 = 47\Omega$, $R_{parasitic} = 100\Omega$ when a) $R_1 = 300\Omega$ and b) $R_1 = 3k\Omega$. Set the amplitude of the voltage source V_1 such that it is a square wave with an amplitude that varies from 2 V to 6 V. The pulse width should be 0.25 ms and the period should be 0.5 ms.

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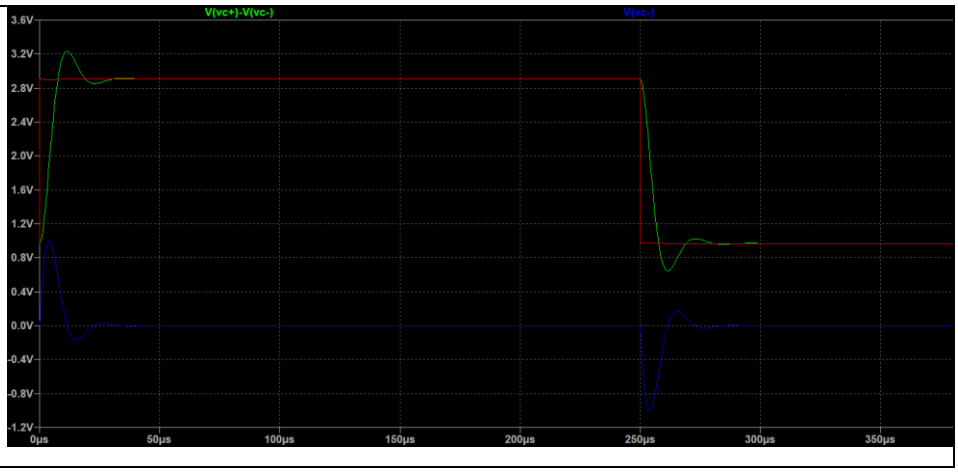
Insert the resulting schematic	a)	
	b)	

2. Graph the voltage across C_1, the voltage across R_1, and the voltage across R_2 for one period of the square wave. Either scale the curves so that the magnitude and shape of each signal is readable or add plots to obtain a stack of graphs of the signals.

Insert the resulting graph with appropriate annotations, units and legends.	a)	

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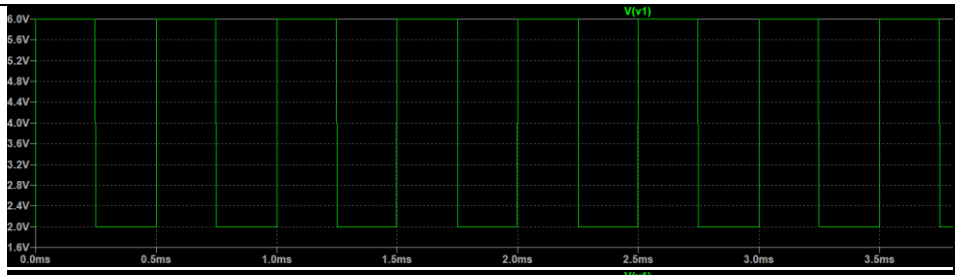
b)



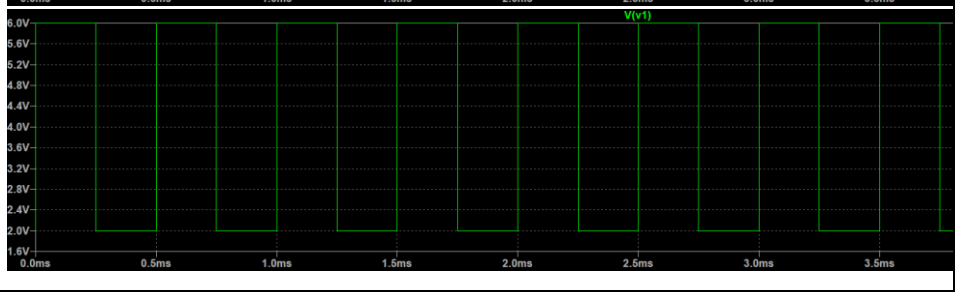
3. Graph the current flowing out of V_1 to show that the voltage source is always a power generating element in the circuit.

Insert the resulting graph with appropriate annotations, units and legends.

a)



b)



4. Measure the period of the underdamped response of the voltage across the capacitor when a) $R_1 = 300\Omega$ and b) $R_1 = 3k\Omega$. Calculate the damped frequency ω_d using the period measured. Determine the percent difference between the expected value of ω_d calculated in Step 1 and the damped frequency calculated from the data graphed in PSpice. Don't forget units.

a) $R_1 = 300\Omega$

$T_{oscillation}$	$T_{\text{oscillation}} = \frac{150.74 - 9.56}{7} = 20.17\mu s$
$f_{oscillation}$	$f_{\text{oscillation}} = \frac{1}{T_{\text{oscillation}}} = \frac{1}{20.17} = 5 \times 10^4 \text{Hz}$
ω_d	$\omega_d = 2\pi f = 3.14 \times 10^5 \text{rad/s}$

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Percent difference between theory and simulation	$\frac{3.14 \times 10^5 - 3.15 \times 10^5}{3.15 \times 10^5} \times 100\backslash\% = 0.32\backslash\%$

b) $R_1 = 3k\Omega$

$T_{oscillation}$	$T_{\text{oscillation}} = 26.84 - 4.04 = 22.8\mu s$
$f_{oscillation}$	$f_{\text{oscillation}} = \frac{1}{T_{\text{oscillation}}} = \frac{1}{22.8} = 4.4 \times 10^4 \text{Hz}$
ω_d	$\omega_d = 2\pi f = 2.8 \times 10^5 \text{rad/s}$
Percent difference between theory and simulation	$\frac{2.8 \times 10^5 - 2.75 \times 10^5}{2.75 \times 10^5} \times 100\% = 1.8\%$

3. Conclusion

1. To reduce the length of time that ringing of the voltage across R_1 can be observed:
   a. Should α increase or decrease?

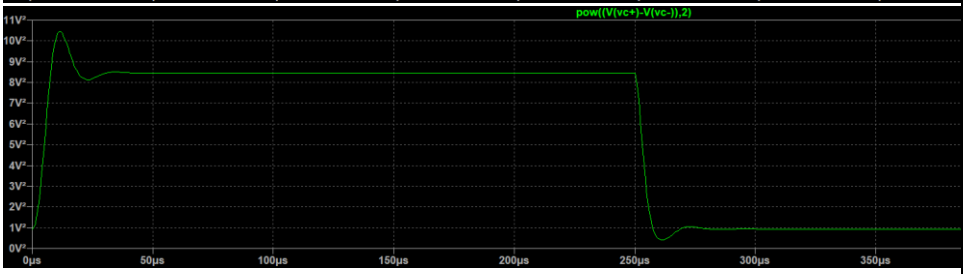
α should	Increase
Justify your answer (text, simulation, and/or equations)	From the simulation we can see that greater α leads to shorter damping time

   b. If R_1 is held constant, should the value of L_1 increase or decrease to reduce the transient response?

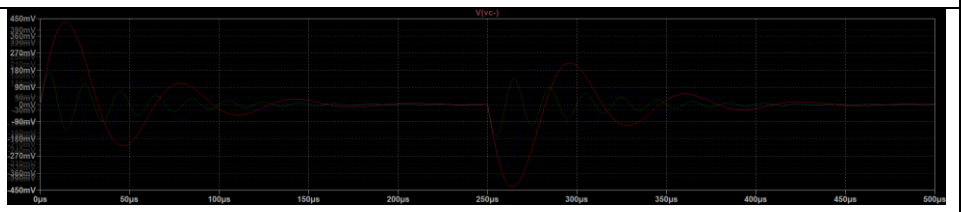
L_1 should	Decrease
Justify your answer (text, simulation, and/or equations)	Since $\alpha = \frac{R_{eq}}{2L}$, and according to a, a greater α is needed to reduce the transient response, so L1 has to decrease.

2. Do you expect that the energy stored in the capacitor will dissipate faster as R_1 increases? Why or why not? Simulate the power across R1 and comment for R1 = 3kΩ and R1 = 300Ω.

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The energy	The energy stored in the capacitor will dissipate faster as R1 increases.
Justify your answer (text, simulation, and/or equations)	Since $E = \frac{1}{2}CU^2$, and C is constant, we can compare the two situations by showing V_{C1}^2 a)  b)  Figures above proves the statement.

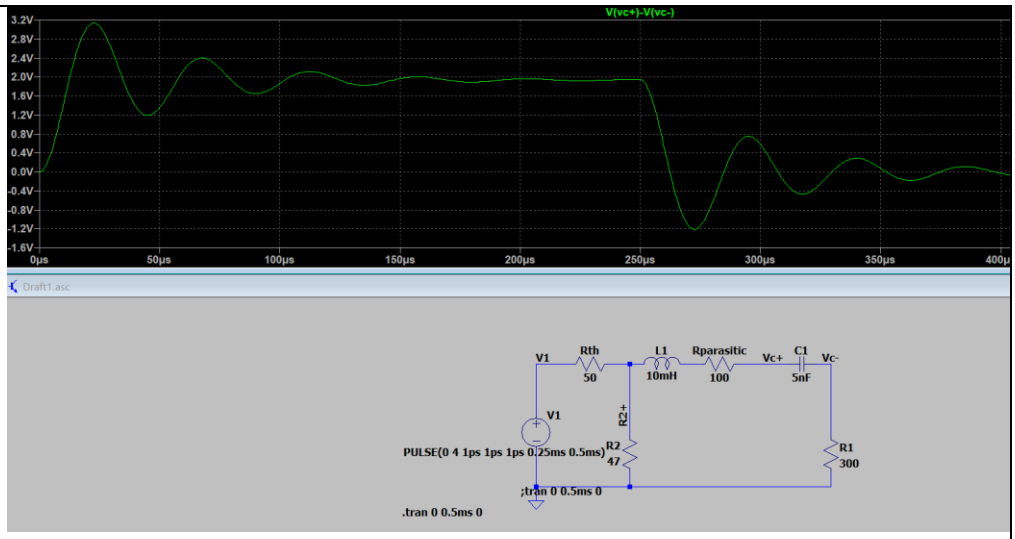
3. Given the relationship between energy, charge, voltage, and capacitance, would you expect that the amplitude of the voltage overshoot in the transient response to increase or decrease if the capacitance of C_1 is increased while the magnitudes of the other components in the circuit are held constant? Simulate a transient response through R1 before answering.

The overshoot	The overshoot will increase as the capacitance increases
Justify your answer (text, simulation, and/or equations)	 The green, blue, red line are V_{R1} when C1 is 1nF , 5nF, 10nF respectly, which proves the statement.

4. If the DC offset is set to 0 V, why would it be inappropriate to use an electrolytic capacitor for C_1 ?

Electrolytic capacitor	Electrolytic capacitor could be easily burnt when the voltage across it is negative.

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Justify your answer	 The simulation shows that there are moments when voltage across C1 becomes negative, when pulse of V1 is set to 0V to 4V.